

AI-DRIVEN HYPERTENSION TEST SYSTEM

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DECLARATION OF ORIGINAL WORK

We hereby declare that this project report, titled Hypertension Project, is our own original work submitted for the **Business Analysis 3.2 (AIBUY3A)** module. All sources of information, references, and assistance used in completing this work have been properly cited and acknowledged. We have not plagiarized any work, nor have submitted someone else's work as our own.

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TABLE OF CONTENTS

1. Introduction----------:4

2. Problem Background and Theme Relevance----------:4-5

3. Problem definition----------:5-6

3.1. What is the problem

3.2. Local municipality and community benefit

4. System Aims and business objectives----------:6-7

4.1. Main Aim

4.2. Business objectives and success criteria

4.3. Target users

4.4 Requirements, Constraints, Risks, and Tools & Techniques Assessment

5. Machine learning specification----------:7-8

5.1. Dataset preparation and feature engineering

5.2. Supervised learning approach and algorithms

6. Theoretical extension----------:8-9

6.1. Time series analysis

6.2. Solution Improvement Techniques

6.3. Natural Language Processing (NLP) and speech recognition

6.4. Deep Learning Integration

6.5. Conversational Chatbot / Softbot Interface

7. Dataset and data used----------:9-10

8. Model evaluation----------:11

9. Solution techniques and improvement----------:11

10. Final practical solution----------:11

11. Project poster----------:12

12. Grammarly results/certificate----------:13

1.Introduction

Artificial Intelligence (AI) is increasingly being used to support decision-making and solve problems in different fields, including healthcare. Machine learning, which is a branch of AI, enables computer systems to learn patterns from data and use those patterns to make predictions.

Hypertension, commonly known as high blood pressure, is a health condition in which blood pressure remains higher than normal. It is an important healthcare concern because it can increase the risk of serious health complications if it is not identified and managed appropriately.

The AI-Driven Hypertension Test System project aims to develop a machine-learning-based system that uses relevant health data to estimate whether a person may fall into a defined blood-pressure or hypertension-related category. The system is intended to demonstrate how AI can support preliminary risk assessment and healthcare decision-making.

The project involves collecting and preparing a suitable dataset, training machine-learning models, evaluating their performance and implementing a practical Python-based solution. The final system's functionality and performance will be documented using the actual results obtained during development.

The system is intended to support, rather than replace, professional medical assessment. Its predictions will not be treated as a confirmed medical diagnosis.

2. Project Background & Theme Relevance

Developing and underdeveloped regions continue to fall behind when it comes to using Fourth Industrial Revolution technologies to solve everyday challenges faced by communities. To unleash the power of 4IR in line with the theme” an AI solution for industries” this project introduces a Python-based Artificial Intelligence solution for the Healthcare Industry. The proposed system applies supervised machine learning classification to perform fast, early screening for hypertension risk using five

accessible health and lifestyle indicators, helping healthcare workers and local pharmacies to bridge the diagnostic gap in resource-constrained environments.

3. Problem Definition

3.1 What is the Problem?

Hypertension, or high blood pressure, is a serious health problem that often develops without noticeable symptoms. Because people may not realise that they have high blood pressure, it can go undetected and untreated for a long time, increasing the risk of serious health problems such as heart disease, stroke, and kidney failure. Traditional blood pressure screening usually requires medical equipment, trained healthcare professionals, and a visit to a healthcare facility. This can be difficult for people who live in underserved areas or have limited access to healthcare services. In addition, a person's risk of developing hypertension can be influenced by several factors, including age, body measurements, lifestyle habits, and existing health conditions. Considering all these factors manually can be difficult and time-consuming. There is therefore a need for a simple and accessible automated tool that can use multiple factors to identify a person's potential risk of hypertension and provide an initial screening before formal medical testing.

3.2 Local Municipality and Community Benefit

Creating this AI-driven screening tool could benefit local municipalities and public healthcare facilities by helping reduce overcrowding at primary healthcare clinics. The tool could be used at local pharmacies, community health centres, and mobile health outreach programmes to quickly identify people who may be at a higher risk of hypertension. These individuals could then be encouraged to undergo proper clinical testing and diagnosis.

By identifying potential risks earlier, the tool could help healthcare facilities make better use of their available resources, increase awareness about hypertension, and support preventative healthcare. It could also provide useful data that municipalities can use

to better understand hypertension risk within their communities and improve public health planning.

4. System Aims and Business Objectives

4.1 Main Aim

The main aim of this project is to develop a Python-based machine learning application that can predict whether a person is at a higher or lower risk of high blood pressure. The prediction will be based on multiple health factors: age, salt intake, smoking status, BMI, family history, BP history, Stress score, Sleep duration, Medication and Exercise Level.

4.2 Business Objectives and Success Criteria

Risk Classification: The system should classify users into two simple categories: Yes or NO for the HAS HYPERTENSION column.

Efficiency: The system should provide the results within two seconds, allowing it to be used for quick preliminary screening.

Accessibility: The system should use information that can be easily attained, variables such as age, salt intake, smoking status, sleep duration can be provided by the patient, and the rest will be tested.

4.3 Target Users

Primary Users: Local healthcare clinics, retail pharmacies, and community health centres that perform initial patient screening.

Secondary Users: Individuals who want to gain an early understanding of their potential risk of high blood pressure.

4.4 Requirements, Constraints, Risks, and Tools & Techniques Assessment

Functional & Non-Functional Requirements: The system must accept 11 health and lifestyle inputs, process them via a trained Random Forest pipeline, and output a binary risk score within 2 seconds.

Constraints: The system is designed for preliminary screening only and does not replace formal clinical diagnosis; input features rely on user disclosure and standard measurements.

Risks & Mitigation: Mitigated risk of data leakage by excluding target-derived fields (e.g., historical diagnosis fields) during training; addressed overfitting by implementing cross-validation and standard feature scaling.

Initial Tools & Techniques Assessment: Python was selected for data processing (Pandas, NumPy) and machine learning modelling (scikit-learn) due to its robustness, rich library ecosystem, and ease of deployment.

5.Machine Learning Specification & Architecture

5.1 Dataset Preparation and Feature Engineering

The model will use a cleaned dataset containing five key health and lifestyle factors. These factors were selected because they can be collected without requiring specialised medical equipment, making the system more suitable for preliminary screening.

The eleven features used by the model are:

1. **Age:** The person's age in years.
2. **Smoking Status:** Indicates whether the person is a smoker.
3. **Body Mass Index (BMI):** A measurement based on a person's height and weight.
4. **Salt Intake:** A measure of the amount of salt or sodium the person consumes.
5. **Stress Score:** A rating showing the individual's perceived level of stress.
6. **BP History:** A record indicating whether the person has a history of hypertension.
7. **Sleep Duration:** Measurement of how long the person sleeps on average.
8. **Medication:** An indicator of whether the person is currently taking prescribed medications.
9. **Family History:** An input checking whether a person has had hypertension in their family.

10. **Exercise Level:** A classification of how frequently and intensely the person engages in physical activity.

5.2 Supervised Learning Approach and Algorithms

The project will use supervised machine learning to classify users into either a likely high blood pressure risk group or a likely low blood pressure risk group. Two classification algorithms will be used and compared:

Primary Algorithm — Logistic Regression: Logistic Regression will be used as the main model because it is suitable for binary classification and provides probability-based predictions that are relatively easy to interpret.

Comparison Algorithm — Decision Tree Classifier: A Decision Tree will be used as a second model to compare its performance with Logistic Regression. It can identify relationships between different input factors and is capable of handling non-linear patterns in the data.

We will also train other supervised models to check if they might be more suitable than the above mentioned.

6. Theoretical Extensions

6.1 Time Series Analysis

In the future, the system could be expanded to track a patient's health information over time. Time-series methods, such as ARIMA or rolling averages, could be used to monitor changes in factors such as BMI or recorded blood pressure across different visits. This could help healthcare providers identify changes in a patient's health and encourage earlier follow-up when necessary.

6.2 Solution Improvement Techniques

The model could also be improved over time by regularly retraining it using new, anonymised patient data. As more verified outcomes become available, the model could learn from this additional information and potentially improve its performance for different populations and local health trends.

6.3 Natural Language Processing (NLP) and Speech Recognition

A voice-based input option could be added to make the system easier to use for people who have difficulty reading or using a traditional form. Users could answer the screening questions verbally, with speech recognition converting their answers into the information required by the machine learning model. Support for different local languages could also make the system more accessible.

6.4 Deep Learning Integration

If the system is expanded in the future to use a much larger dataset with more health-related information, deep learning models such as Deep Neural Networks (DNNs) or Multi-Layer Perceptions (MLPs) could be considered. These models may be useful for identifying more complex relationships between multiple health factors that are not captured by the current five-feature model.

6.5 Conversational Chatbot / Softbot Interface

Another possible improvement would be to add a chatbot or softbot interface through a platform such as a web portal or WhatsApp. The chatbot could guide users through a short questionnaire, collect the five required inputs, send the information to the Python machine learning model, and return the predicted risk category. This would provide a simple and convenient way for users to interact with the screening system.

7. Machine Learning

7.1 Data importing and Preprocessing.

The structured raw dataset (hypertension_dataset.csv) was imported into a Pandas DataFrame called ‘df’.

Inspected the dataset structure, summary statistics and missing values which resulted in replacing the missing numerical values in their own column.

7.2 Feature and Target Separation.

A target variable was defined as ‘Has_Hypertension’ which was then converted into binary values. (1 = Yes, 0 = No).

Filtered inputs features (X), which did exclude non-numerical columns and fields like ‘BP_History’ and ‘Medication’ to prevent data from leaking.

7.3 Train-Test Split

Split Ratio was done which allocated 80% to model training and the remaining 20% for testing.

Applied random_state = 42 for reproducibility and stratification to maintain class distribution across splits.

7.4 Algorithm Selection & Feature Scaling

Selected Classifiers:

1. Logistic Regression (Linear binary classification)
2. Decision Tree (Rule-based classification)
3. Random Forest (Ensemble tree model)
4. K-Nearest Neighbours (KNN) (Distance-based instance learning)
5. Support Vector Machine (SVM) (Boundary hyperplanes)

Scaling: Applied StandardScaler to scale features for scale-sensitive models (Logistic Regression, KNN, SVM).

7.5 Model Training

Trained all 5 classifiers using identical training data ([Equation]) to ensure a fair comparative benchmark.

7.6 Model Testing and Performance Evaluation

Evaluated trained models on the 20% unseen test dataset (Xtest*Xtest*) to generate hypertension risk predictions.

Evaluated classifiers across standard metrics:

- **Accuracy & F1-Score:** Measured overall correctness and balanced performance.
- **Precision & Recall:** Assessed false-positive rates vs. true-positive detection.
- **Confusion Matrix:** Tracked True Positives, True Negatives, False Positives, and False Negatives.
- **ROC Curve & AUC:** Evaluated class distinction boundaries.

8. Model Evaluation

Evaluated models using a Random Forest Classifier (RandomForestClassifier) integrated within a scikit-learn pipeline to ensure robust classification performance. Continuous tracking of precision and recall metrics ensures patient risk categories remain clinically reliable before presenting screening feedback.

9. Solution Techniques and Improvement

9.1

Upgraded the pipeline from simple baseline models to an ensemble Random Forest approach combined with feature scaling (StandardScaler) to better capture non-linear health interactions. The pipeline allows for dynamic model enhancement over time by retraining fresh, anonymized user screening data to adapt to shifting community health trends.

9.2 Tools, Scripts, and Technologies Used for Training & Improvement:

Python Libraries: Utilized Pandas and NumPy for data manipulation and numerical operations; scikit-learn for pipeline creation, model training, feature scaling, and evaluation metrics.

Development Scripts & Modules: Managed distinct workflow modules (Importing_dataset.py, seperating.py, suitable_algorithms.py, training_&_testing.py, and train_model.py) to handle end-to-end data processing and model tuning.

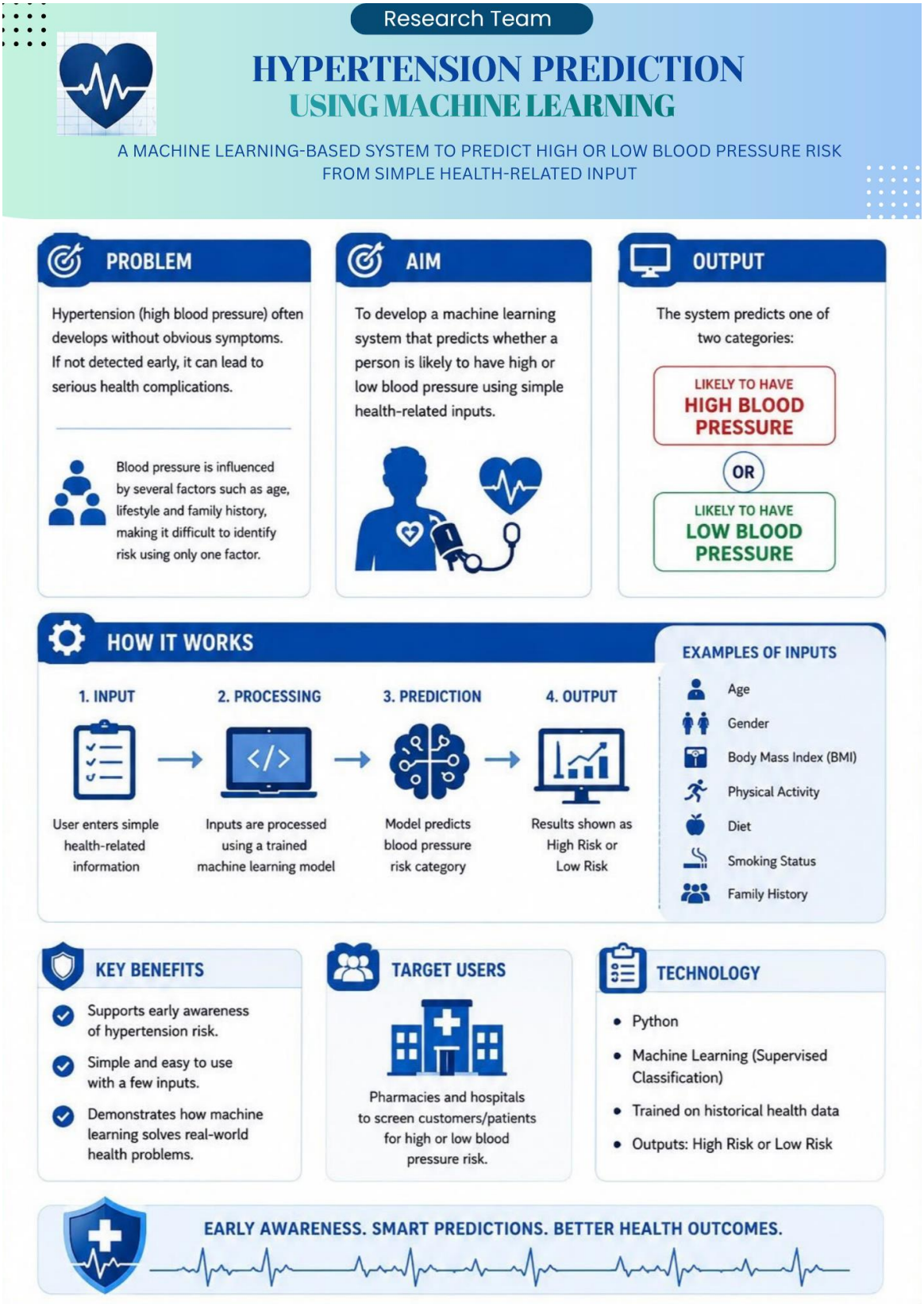
Model Serialization & Version Control: Saved and serialized the optimized model file (hypertension_rf_model.pkl) for fast inference; tracked project assets and version control via Git and GitHub.

10.Final Practical Solution

Deployed the serialized Random Forest pipeline to enable real-time risk predictions.

Future Use for Users: Allows end-users, local pharmacies, and community clinics to input lifestyle indicators (Age, BMI, Stress, Sleep, etc.) into a web portal or chatbot interface to receive instant preliminary hypertension risk assessments.

Project poster



Grammarly results

Performance

Text score: 90 out of 100. This score represents the quality of writing in this document. You can increase it by addressing Grammarly's suggestions.



Word Count

Characters	11141	Reading time	6m 22s
Words	1531	Speaking time	12m 45s
Sentences	133		

Readability

Metrics compared to other Grammarly users

Word length	5.6	Above average
Sentence length	11.5	Above average
Readability score	36	

Your text is likely to be understood by a reader who has at least some college education, but it may not be easy to read.

Vocabulary

Metrics compared to other Grammarly users

Unique Words	36%	Above average
Rare Words	35%	Above average